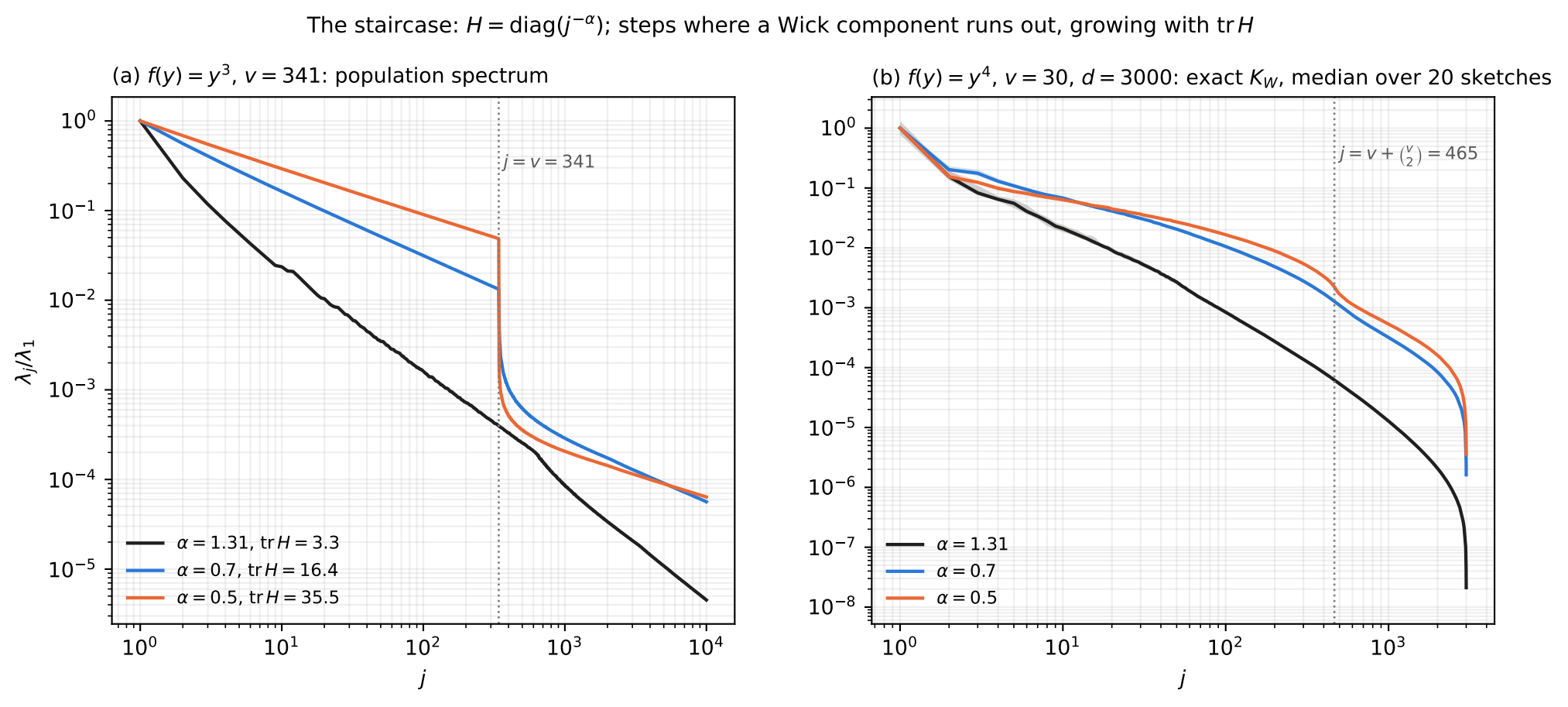

The staircase: $H = \mathrm{diag}(j^{-\alpha})$; steps where a Wick component runs out, growing with $\mathrm{tr}\,H$
(a) $f(y) = y^3$, $v = 341$: population spectrum
λ_j/λ_1
j
$j = v = 341$
$\alpha = 1.31$, $\mathrm{tr}\,H = 3.3$
$\alpha = 0.7$, $\mathrm{tr}\,H = 16.4$
$\alpha = 0.5$, $\mathrm{tr}\,H = 35.5$
(b) $f(y) = y^4$, $v = 30$, $d = 3000$: exact K_W, median over 20 sketches
j
$j = v + \binom{v}{2} = 465$
$\alpha = 1.31$
$\alpha = 0.7$
$\alpha = 0.5$